

(1001CJA100220019)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
06-09-2020
JEE(Main + Advanced) : NURTURE COURSE [PHASE : PHASE-TNPS, TOPS & TRPS]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,C	A,B,D	A,B,C,D	A,B,C,D	A,B,C,D	A	A	C	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	1.00	5.00	5.00	5.00	5.00	4.00	4.00	5.00		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	A,B,D	A,C	A,B,D	A,C,D	A,B	B	D	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	3.00	5.00	1.50	6.00	8.00	5.00	87.69	4.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,B,C	A,B,C,D	A,B,C	B,C	B,C	C	B	C	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	1.73 or 1.74	1024.00	0.80	0.78 or 0.79	0.41 or 0.42	2.25	4.00	1.00		

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JEE(Main + Advanced) : NURTURE COURSE [PHASE : PHASE-TNPS, TOPS & TRPS]**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,C	A,B,D	A,B,C,D	A,B,C,D	A,B,C,D	A	A	C	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	1.00	5.00	5.00	5.00	5.00	4.00	4.00	5.00		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B	A,B,D	B,C,D	B,C,D	A,D	A,C	B	A	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	5.00	10.8	4.54	4.00	2.00	2.00	22.40	30.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	B,D	A,B	A,C	A	B	B	A	D	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	4.90	6.00	2.75	0.50	0.00	0.50	0.25	1.50		

JEE(Main + Advanced) : NURTURE COURSE [PHASE : PHASE-TNPS, TOPS & TRPS]**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A,C)**

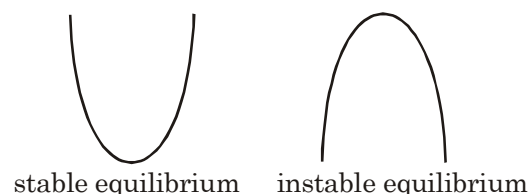
Sol. $F = \frac{-\partial u}{\partial x}$

At point B $\Rightarrow \frac{\partial u}{\partial x} < 0 \Rightarrow F > 0$

At point C $\frac{\partial u}{\partial x} = 0 \Rightarrow F = 0$

At equilibrium $\Rightarrow \frac{\partial u}{\partial x} = 0$

acceleration and force have direction.

2. **Ans. (A, C)**

Sol. $W = \vec{F} \cdot \vec{r}$

$36 = 3C + 8 + C^2$

solving we get, $C = 4, -7$

3. **Ans. (A,B,D)**

Sol. $a_t = \frac{dv}{dt}$

$a_c = \frac{v^2}{R}$

$a = \sqrt{a_c^2 + a_t^2}$

4. **Ans. (A, B, C, D)**

Sol. $\omega = \frac{2\pi}{T} = \frac{2\pi}{60} = \frac{\pi}{30} \text{ rad/sec}$

$v = r\omega = 1 \times \frac{\pi}{30} = \frac{\pi}{30} \text{ m/sec}$

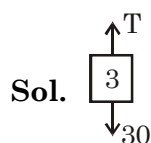
$= \frac{\pi \times 100}{30} \text{ cm/sec} = \frac{10\pi}{3} \text{ cm/sec}$

$a_c = r\omega^2 = 100 \text{ cm} \times \frac{\pi^2}{30 \times 30} = \frac{\pi^2}{9} \text{ cm/sec}^2$

5. **Ans. (A,B,C,D)**

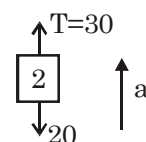
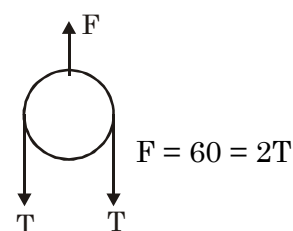
Sol. $a = \frac{40}{8} = 5 \text{ m/s}^2$

$T = \frac{40(x)}{100} \text{ N}$

6. **Ans. (A,B,C,D)**

velocity constant so acceleration zero

$T = 30$



$30 - 20 = 2a$

$a = 5$

7. **Ans. (A)**

Sol. Will move together with acceleration of 3 m/s^2

$f_s = 6 \text{ N}$

8. **Ans. (A)**

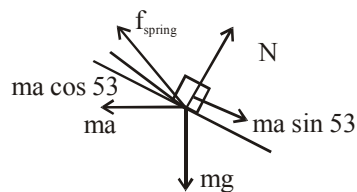
Sol. A & B will move separately & friction will be kinetic.

9. **Ans. (C)**

10. Ans. (B)

Sol. w.r.t wedge

$$\text{If } ma \cos 53 = mg \sin 53$$



$$a = \frac{4g}{3}$$

spring in natural length

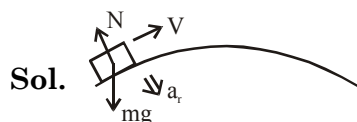
SECTION-II

1. Ans. 1.00

$$\text{Sol. } P = \frac{mgh}{T} = \frac{5 \times 60 \times 10 \times 20}{60} = 1000W$$

$$[\because m = 5 \times 60kg; h = 20m \text{ and } T = 60 \text{ sec}]$$

2. Ans. 5.00



Sol.

$$mg \cos \alpha - N = \frac{mv^2}{R}$$

$$\therefore v = \sqrt{gR \cos \alpha}$$

3. Ans. 5.00

$$\text{Sol. } \vec{F} = mg \sin \theta$$

$$\vec{d} = 1m$$

$$\sin \theta = \frac{0.5}{1} = \frac{1}{2}$$

$$W = 5 \times 1 = 5J$$

4. Ans. 5.00

$$\text{Sol. } a_{\text{of block}} = \frac{3g - 2g}{5} = 2m/s^2$$

$$v = 2 \times \frac{1}{\sqrt{2}} = \sqrt{2}m/s$$

$$\therefore \Delta KE = \frac{1}{2} 2 \times (\sqrt{2})^2 + \frac{1}{2} (3) (\sqrt{2})^2 = 5$$

5. Ans. 5.00

$$\text{Sol. } \mu mg = \frac{mv^2}{R}$$

6. Ans. 4.00

$$\text{Sol. } a_A = \frac{6g - 2g}{2} = 2g$$

7. Ans. 4.00

$$\text{Sol. } a = \frac{30 - 15}{3} = 5m/s^2$$

$$N - 10 = 2 \times 5$$

$$N = 20 \text{ Newton}$$

8. Ans. 5.00

$$\text{Sol. } a_A = \frac{100 - 40}{4} = 15 \text{ m/s}^2;$$

$$a_B = \frac{100 - 50}{5} = 10 \text{ m/s}^2$$

$$\Rightarrow a_A - a_B = 15 - 10 = 5 \text{ m/s}^2$$

PART-2 : CHEMISTRY

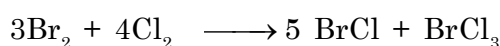
SOLUTION

SECTION-I

1. Ans.(A,B,C)
2. Ans.(A,B,D)
3. Ans. (A,C)
4. Ans. (A,B,D)
5. Ans.(A,C,D)
6. Ans. (A, B)
7. Ans.(B)
8. Ans.(D)
9. Ans. (C)
10. Ans. (A)

SECTION-II

1. Ans.(3.00)
2. Ans.(5.00)
3. Ans.(1.50)
4. Ans.(6.00)
5. Ans (8.00)
6. Ans. (5.00)
7. Ans.(87.69)
8. Ans.(4.00)



PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A,C,D)

Sol. Let $S = \sum_{k=1}^{100} \sin(kx) \cos(101-k)x$

$$\Rightarrow S = \sin x \cos 100x + \sin 2x \cos 99x + \dots + \sin 100x \cos x \quad \dots (i)$$

$$S = \cos x \sin 100x + \cos 2x \sin 99x + \dots + \sin x \cos 100x \quad \dots (ii)$$

(on writing in reverse order)

Adding, we get

$$2S = \underbrace{\sin 101x + \sin 101x + \dots + \sin 101x}_{100 \text{ times}}$$

$$\text{Hence } S = \frac{100}{2} \sin(101x) = 50 \sin(101x) \text{ Ans.}$$

2. Ans. (A,B,C)

Sol. (A) $(\tan \theta + \cot \theta)^2$

$$= \tan^2 \theta + \cot^2 \theta + 2$$

$$= (1 + \tan^2 \theta) + (1 + \cot^2 \theta)$$

$$= \sec^2 \theta + \operatorname{cosec}^2 \theta$$

$$= \frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}$$

$$= \sec^2 \theta \operatorname{cosec}^2 \theta$$

(B) As $3\theta = 2\theta + \theta$

$$\therefore \tan 3\theta = \tan (2\theta + \theta)$$

$$\tan 3\theta = \frac{\tan 2\theta + \tan \theta}{1 - \tan 2\theta \tan \theta}$$

$$\Rightarrow \tan 3\theta - \tan 2\theta - \tan \theta = \tan 3\theta \tan 2\theta \tan \theta$$

(C) $\frac{1 + \cot^2 \theta}{1 - \sin^2 \theta} = \frac{\operatorname{cosec}^2 \theta}{\cos^2 \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cos^2 \theta}$

$$= \sec^2 \theta + \operatorname{cosec}^2 \theta$$

(D) $\tan \left(\frac{\pi}{4} + \theta \right) + \tan \left(\frac{\pi}{4} - \theta \right) = \left(\frac{1 + \tan \theta}{1 - \tan \theta} \right)$

$$+ \left(\frac{1 - \tan \theta}{1 + \tan \theta} \right) = \frac{(1 + \tan \theta)^2 + (1 - \tan \theta)^2}{1 - \tan^2 \theta}$$

3. Ans. (A,B,C,D)

Sol. $\sum_{r=1}^n r(2r^2 + 7r + 6) = \sum_{r=1}^n (2r^3 + 7r^2 + 6r)$

$$= \frac{n^2(n+1)^2}{2} + \frac{7n(n+1)(2n+1)}{6} + 3n(n+1)$$

$$= \frac{n^2(n^2 + 2n + 1)}{2} + \frac{7n(2n^2 + 3n + 1)}{6} + 3n^2 + 3n$$

$$= \frac{1}{2} n^4 + \frac{10}{3} n^3 + 7n^2 + \frac{25}{6} n$$

4. Ans. (A,B,C)

Sol. We have $a_9 = 5$

$$\Rightarrow a_1 + 8d = 5$$

$$\Rightarrow 3 + 8d = 5$$

$$\therefore d = \frac{1}{4} = \text{common difference of A.P.}$$

Also, $g_9 = 5$

$$\Rightarrow g_1 r^8 = 5 \Rightarrow 2r^8 = 5$$

$$\therefore r = \left(\frac{5}{2}\right)^{\frac{1}{8}} = \text{common ratio of G.P.}$$

Now

$$(A) a_5 g_{17} = \left(3 + 4 \times \frac{1}{4}\right) \left(2 \times \left(\frac{5}{2}\right)^{\frac{16}{8}}\right)$$

$$= (4) \left(2 \times \frac{25}{4}\right) = 50$$

$$(B) a_{21} g_{25} = \left(3 + 20 \times \frac{1}{4}\right) \left(2 \times \left(\frac{5}{2}\right)^{\frac{24}{8}}\right)$$

$$= (8) \left(2 \times \frac{125}{8}\right) = 250$$

$$(C) a_{29} g_9 = \left(3 + 28 \times \frac{1}{4}\right) \left(2 \times \left(\frac{5}{2}\right)^{\frac{8}{8}}\right)$$

$$= (10) \left(2 \times \frac{5}{2}\right) = 50 = a_5 g_{17}$$

$$(D) a_{37} g_{17} = \left(3 + 36 \times \frac{1}{4}\right) \left(2 \times \left(\frac{5}{2}\right)^{\frac{16}{8}}\right)$$

$$= (12) \left(2 \times \frac{25}{4}\right) = 150$$

5. Ans. (B,C)

Sol. We have $2 \cos \frac{A+B}{2} \cos \frac{A-B}{2} = 4 \sin^2 \frac{C}{2}$

$$\text{or } \cos \frac{A-B}{2} = 2 \sin \frac{C}{2} \left(\sin \frac{C}{2} = \cos \frac{A+B}{2} \right)$$

$$\text{or } \cos \frac{A-B}{2} = 2 \cos \frac{A+B}{2}$$

$$\text{or } \cos \frac{A-B}{2} - \cos \frac{A+B}{2} = \cos \frac{A+B}{2}$$

$$2 \sin \frac{A}{2} \sin \frac{B}{2} = \cos \frac{A}{2} \cos \frac{B}{2} - \sin \frac{A}{2} \sin \frac{B}{2}$$

$$3 \sin \frac{A}{2} \sin \frac{B}{2} = \cos \frac{A}{2} \cos \frac{B}{2}$$

$$\text{or } \tan \frac{A}{2} \tan \frac{B}{2} = \frac{1}{3}$$

$$\text{Now } \frac{\Delta}{s(s-a)} \cdot \frac{\Delta}{s(s-b)} = \frac{1}{3}; \frac{s-c}{s} = \frac{1}{3}$$

$$\therefore 2s = 3c$$

$$a + b + c = 3c$$

$$\Rightarrow a + b = 2c$$

$$\Rightarrow a, c, b \text{ are in A.P.}$$

6. Ans. (B,C)

$$\text{Sol. } D = \begin{vmatrix} 2 & \lambda & 6 \\ 1 & 2 & \mu \\ 1 & 1 & 3 \end{vmatrix} = 2(6 - \mu)$$

$$-\lambda(3 - \mu) + 6(1 - 2)$$

$$= 6 - 2\mu - 3\lambda + \lambda\mu$$

$$= (\lambda - 2)(\mu - 3)$$

$$Dx = \begin{vmatrix} 8 & \lambda & 6 \\ 5 & 2 & \mu \\ 4 & 1 & 3 \end{vmatrix}$$

$$= 8(6 - \mu) - \lambda(15 - 4\mu) + 6(5 - 8)$$

$$= 30 - 15\lambda - 8\mu + 4\lambda\mu$$

$$= (4\mu - 15)(\lambda - 2)$$

$$D_y = \begin{vmatrix} 2 & 8 & 6 \\ 1 & 5 & m \\ 1 & 4 & 3 \end{vmatrix} = 2(15 - 4\mu) - 8(3 - \mu) + 6(4 - 5)$$

$$= 0$$

$$D_z = \begin{vmatrix} 2 & \lambda & 8 \\ 1 & 2 & 5 \\ 1 & 1 & 4 \end{vmatrix} = 2(8 - 5) - \lambda(4 - 5) + 8(1 - 2)$$

$$= 6 + \lambda - 8$$

$$= \lambda - 2$$

for no solution

$D = 0$ and atleast one of

$$D_x, D_y, D_z$$

should be non zero

clearly $\lambda \neq z$ and $\mu = 3$

Paragraph for Question 7 and 8

7. Ans. (C)

8. Ans. (B)

$$\text{Sol. } \Delta = a \begin{vmatrix} 1 & a & 0 \\ 1 & 2a+b & (a+b)^2 \\ 0 & 1 & 2a+3b \end{vmatrix}$$

$$R_2 \Rightarrow R_2 - R_1$$

$$= a \begin{vmatrix} 1 & a & 0 \\ 0 & a+b & (a+b)^2 \\ 0 & 1 & 2a+3b \end{vmatrix}$$

$$= a(a+b) \begin{vmatrix} 1 & a & 0 \\ 0 & 1 & a+b \\ 0 & 1 & 2a+3b \end{vmatrix}$$

$$= a(a+b) (1.(2a+3b-a-b) - 0 + 0)$$

$$= a(a+b) (a+2b)$$

now check options

Paragraph for Question 9 and 10

9. Ans. (C)

$$\text{Sol. } c^2 = a^2 + b^2 - 2ab \cos c$$

$$= 16 + 64 - 64 \frac{1}{2}$$

$$= 16 + 32 = 48$$

$$c = 4\sqrt{3}$$

10. Ans. (C)

$$\text{Sol. } \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$= \frac{64 + 48 - 16}{2 \times 8 \times 4\sqrt{3}}$$

$$= \frac{6}{4\sqrt{3}} = \frac{\sqrt{3}}{2}$$

$$A = \frac{\pi}{6}$$

SECTION-II

1. Ans. 1.73 or 1.74

$$\text{Sol. } \text{We have } \frac{4 + \sec 20^\circ}{\operatorname{cosec} 20^\circ}$$

$$= \frac{\sin 20^\circ}{\cos 20^\circ} (4 \cos 20^\circ + 1) = \frac{2 \sin 40^\circ + \sin 20^\circ}{\cos 20^\circ}$$

$$= \frac{\sin 40^\circ + (\sin 40^\circ + \sin 20^\circ)}{\cos 20^\circ}$$

$$= \frac{\sin 40^\circ + 2 \sin 30^\circ \cos 10^\circ}{\cos 20^\circ} = \frac{\sin 40^\circ + \sin 80^\circ}{\cos 20^\circ}$$

$$= \frac{2 \sin 60^\circ \cos 20^\circ}{\cos 20^\circ} = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$$

Hence square of the value of expression
= 3 Ans.

2. Ans. 1024.00

Sol. We have

$$\frac{2^{10} \sin\left(\frac{5\pi}{2^{11}}\right) \cos\left(\frac{5\pi}{2^{11}}\right) \cos\left(\frac{5\pi}{2^{10}}\right) \cos\left(\frac{5\pi}{2^9}\right) \dots \cos\left(\frac{5\pi}{2^3}\right) \cos\left(\frac{5\pi}{2^2}\right)}{2^{10}}$$

$$= \frac{\sin \frac{5\pi}{2}}{2^{10}} = \frac{\sin 450^\circ}{2^{10}} = \frac{1}{2^{10}} = \frac{1}{1024}$$

$\Rightarrow$ Reciprocal of the value of the product

= 1024 Ans.

3. Ans. 0.80

Sol. $\frac{1}{2} r^2 (\sin \alpha + \sin \beta + \sin \gamma) = 8$ and $\pi r^2 = 20$

4. Ans. 0.78 or 0.79

Sol. The roots of the equation $x^2 + \sqrt{2}x + 1 = 0$ are non real, so both the quadratic equations must be identical.

$$\therefore \frac{a}{1} = \frac{b}{\sqrt{2}} = \frac{c}{1} = k \text{ (let)}$$

$$\Rightarrow a = k; b = \sqrt{2}k; c = k$$

$$\text{Now, } \cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{k^2 + 2k^2 - k^2}{2(k)(\sqrt{2}k)}$$

$$= \frac{1+2-1}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$$

Hence $\angle C = 45^\circ$ Ans.

5. Ans. 0.41 or 0.42

Sol. $\tan \frac{\pi}{8}$ is standard value

$$\tan \frac{\pi}{8} = \sqrt{2} - 1 = 0.414$$

6. Ans. 2.25

Sol. $\frac{\lambda : 1}{A \quad P \quad B}$

$$P = \left(\frac{7\lambda + 8}{\lambda + 1}, \frac{-4\lambda + 9}{\lambda + 1} \right)$$

P is on x-axis

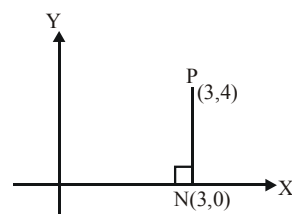
hence

$$\frac{9 - 4\lambda}{\lambda + 1} = 0$$

$$\lambda = \frac{9}{4} = 2.25$$

7. Ans. 4.00

Sol.



$$d = PN = \sqrt{(3-3)^2 + (4-0)^2} = 4$$

8. Ans. 1.00

Sol. $x^4 - 2ax^2 + x + a^2 - a = 0$
 $a^2 - (2x^2 + 1)a + x^4 + x = 0$

$$D = (2x^2 + 1)^2 - 4(x^4 + x) = 4x^4 + 4x^2 + 1 - 4x^4 - 4x = 4x^2 + 1 - 4x$$

$$D = (2x - 1)^2$$

$$a = \frac{(2x^2 + 1) \pm \sqrt{(2x - 1)^2}}{2(1)}$$

$$a = \frac{2x^2 + 1 \pm |2x - 1|}{2}$$

$$2a = 2x^2 + 1 + 2x - 1$$

$$2a = 2x^2 + 2x$$

$$2a = 2x^2 + 1 - 2x + 7$$

$$x^2 + x - a = 0$$

$$D \geq 0$$

$$1 + 4a \geq 0$$

$$4a \geq -1$$

$$a \geq -\frac{1}{4}$$

$$a = x^2 - x + 1$$

$$x^2 - x + 1 - a = 0$$

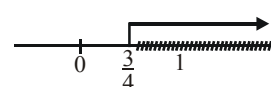
$$1 - 4(1 - a) \geq 0$$

$$1 - 4 + 4a \geq 0$$

$$4a - 3 \geq 0$$

$$4a \geq 3$$

$$a \geq \frac{3}{4}$$



$$a \geq \frac{3}{4} \text{ Hence } a = 1$$

JEE(Main + Advanced) : NURTURE COURSE [PHASE : PHASE-TNPS, TOPS & TRPS]**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I****1. Ans. (B,C)****Sol.** As slope increases from O to A $\Rightarrow$ velocity increasesAs slope is constant from A to B $\Rightarrow$ velocity constantAs slope decreases from B to C $\Rightarrow$ velocity decreases

$$wD = \Delta k$$

As velocity increases $\Rightarrow \Delta k$ increases $\Rightarrow$ and vice versa.**2. Ans. (A, C)**

$$\text{Sol. } a_c = \frac{v^2}{R} \quad a_t = 0$$

$$a_c = \frac{400}{250} \text{ m/s}^2 = 1.6 \text{ m/s}^2$$

Since $a_t = 0$

so frictional force will be always towards centre.

3. Ans. (A,C,D)

$$\text{Sol. } W_{mg} = mgh = 5 \times 10 \times 6 = 300 \text{ J}$$

$$\Delta U = -300 \text{ J}$$

 $W_N = 0$ (displacement is perpendicular to normal)

$$W_f = -\mu mg \cos \theta x = -0.5 \times 5 \times 10 \times \frac{4}{5} \times 10$$

$$= -200 \text{ J}$$

$$W_{\text{all}} = \Delta KE$$

4. Ans. (A,C)

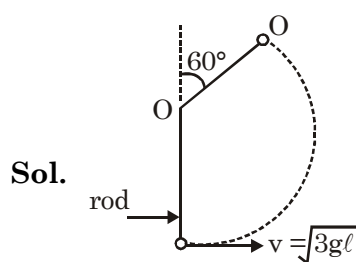
$$\text{Sol. } \frac{dU}{dx} = (x+1)(1) + (x+2)1 = 2x+3$$

$$x = -\frac{3}{2} \text{ is the equilibrium position}$$

$$\frac{d^2U}{dx^2} = 2 > 0 \text{ so equilibrium stable}$$

$$F = -\frac{dU}{dx} = 0 \Rightarrow -2x - 3 = 0$$

$$x = -\frac{3}{2} \text{ Ans.}$$

5. Ans. (B, D)**Sol.**

By applying W.E.T.

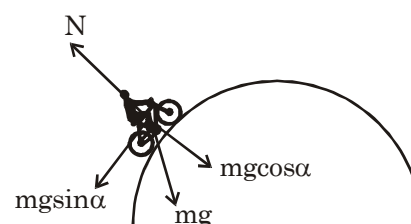
$$0 - \frac{1}{2}mv^2 = -mg\ell(1 + \cos \theta)$$

$$v = \sqrt{3g\ell} \Rightarrow \cos \theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

$$H = \ell + \ell \cos 60^\circ = \frac{3\ell}{2}$$

$$a_T = g \sin 60^\circ = \frac{g\sqrt{3}}{2}$$

6. Ans. (B,D)**Sol.** If speed is constant the friction will have to be upwards for making net force zero
In tangential direction

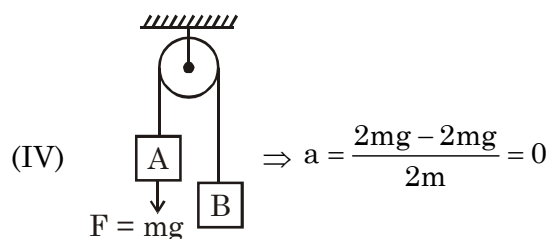
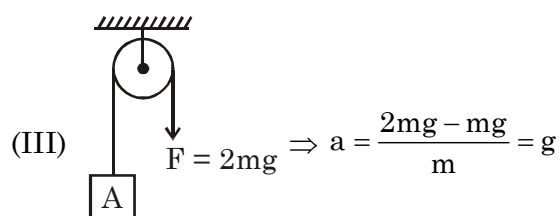
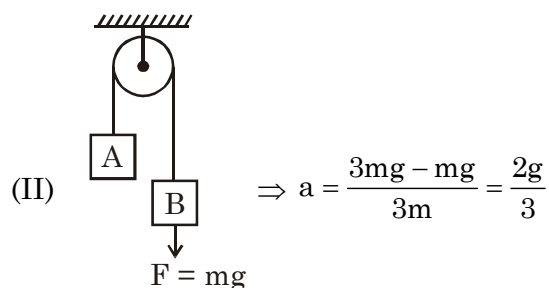
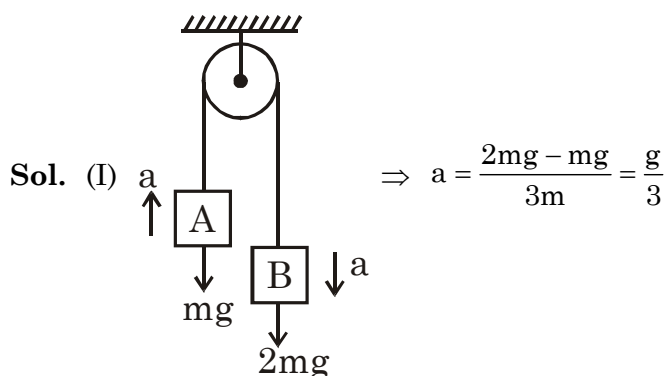
$$mg\cos\alpha - N = \frac{mv^2}{R}$$

$$N = mg \cos \alpha - \frac{mv^2}{2} \text{ (fixed)}$$

Increases while going up.

So normal increases as cyclist goes up.

7. Ans. (C)



8. **Ans. (B)**

Sol. (P) $T = \frac{2u \sin \theta}{g}$

$$T = 8 \text{ sec}$$

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$R = 240 \text{ m}$$

(Q)w.r.t. dog horizontal velocity of ball = 10 m/s

So disp = $10 \times 8 = 80$ m

(R) at $t = 2$

$$\vec{v}_{b/G} = 30\hat{i} + 20\hat{j}$$

$$\vec{V}_{\text{dog/G}} = 20\hat{i}$$

$$|\vec{v}_{b/d}| = |10\hat{i} + 20\hat{j}| = 10\sqrt{5} \text{ m/s}$$

(S) at $t = 5$

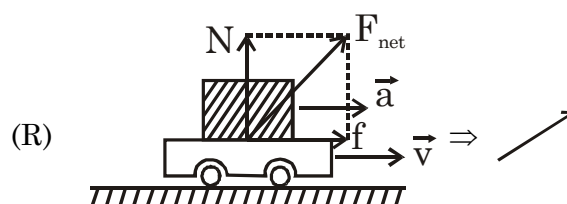
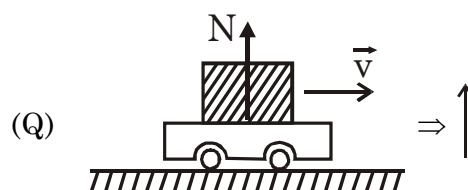
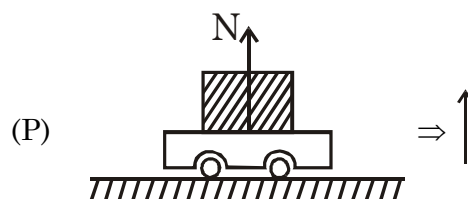
$$\vec{v}_{b/G} = 30\hat{i} - 10\hat{j}$$

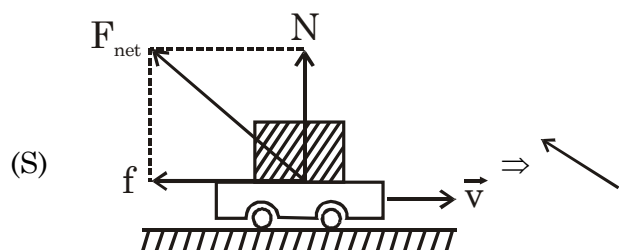
$$\vec{V}_{d/G} = 20\hat{i}$$

$$|\vec{v}_{b/d}| = |10\hat{i} - 10\hat{j}| = 10\sqrt{2} \text{ m/s}$$

9. Ans. (B)

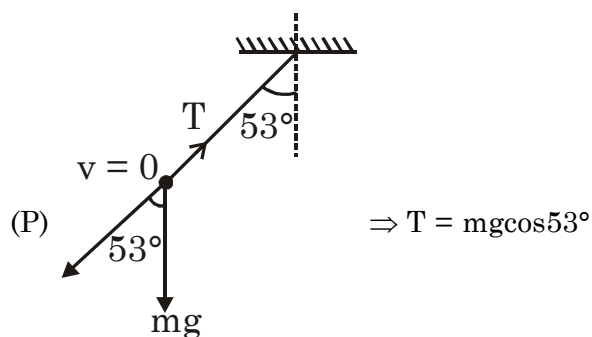
Sol. $\mathbf{F}_{\text{net}} = \vec{\mathbf{N}} + \vec{\mathbf{f}}$



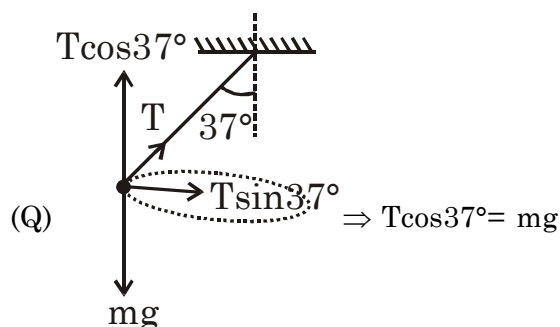


10. Ans. (A)

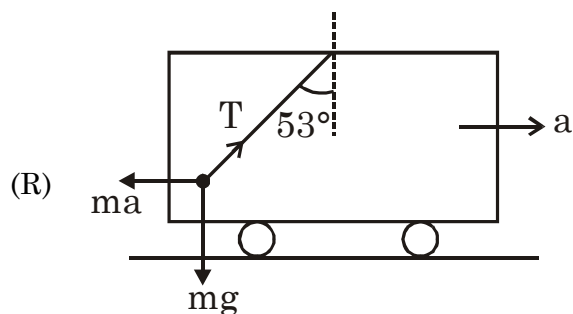
Sol.



$$= \frac{3mg}{5}$$

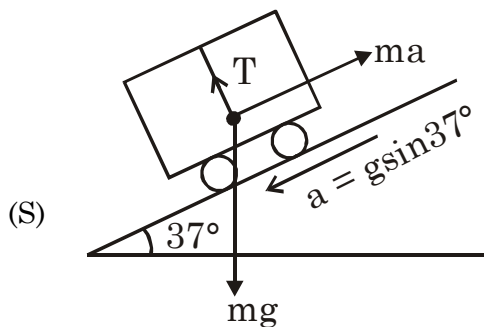


$$\Rightarrow T = \frac{5mg}{4}$$



$$\Rightarrow T \cos 53^\circ = mg \Rightarrow$$

$$T = \frac{5mg}{3}$$



$$\Rightarrow T = mg \cos 37^\circ \Rightarrow T = \frac{4mg}{5}$$

SECTION-IV

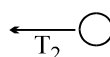
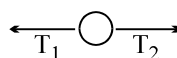
1. Ans. 25.00

Sol. $T_1 \cos 60 = mg \Rightarrow T_1 = 2mg = 20$

$$T_2 = mg \cos 60 \Rightarrow T_2 = \frac{mg}{2} = 5$$

2. Ans. 1.12 to 1.13

Sol. Drawing the FBDs for masses M_1 and M_2



$$T_1 - T_2 = M_1 R_1 \omega^2 \quad T_2 = M_2 R_2 \omega^2$$

$$\therefore \frac{T_1 - T_2}{T_2} = \frac{M_1 \cdot R_1}{M_2 \cdot R_2} = \frac{1}{4} \cdot \frac{1}{2}$$

$$\therefore \frac{T_1}{T_2} = 1 + \frac{1}{8} = \frac{9}{8}$$

3. Ans. 1.00

Sol. When 1kg mass goes down by 1m 2k mass moves by 0.5m and relationship between after velocities in $v_1 = 2v_2$ where v_1 & v_2 are velocities of 1kg and 2kg blocks. Applying work energy theorem for entire system

$$W_{mg} + W_r = \Delta(K.E.)$$

$$m_1 g h_1 - \mu m_2 g x_2 = K.E.$$

$$1 \times 10 \times 1 - 0.7 \times 2 \times 10 \times \frac{1}{2} = \frac{1}{2}(1)v_1^2 + \frac{1}{2}(2)v_2^2$$

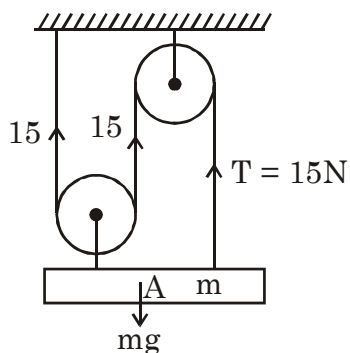
$$10 - 7 = \frac{1}{2}(2v_2)^2 + v_2^2$$

$$3 = 3v_2^2$$

$$v_2 = \sqrt{\frac{3}{3}} = 1 \text{ m/s}$$

4. Ans. 4.50

Sol.



$$45 = mg$$

$$\Rightarrow m = 4.5 \text{ kg}$$

5. Ans. 3.00

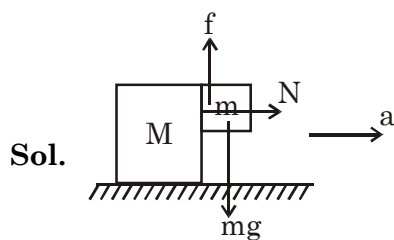
Sol. $F = -\frac{dU}{dr} = -(2x\hat{i} + 2y\hat{j} + 2z\hat{k})$

6. Ans. 10.00

Sol. $Mg - N_A = \frac{mv^2}{R}$

$$N_B - mg = \frac{mv^2}{R}$$

7. Ans. 25.00



Sol.

$$N = ma \dots\dots(1)$$

$$f = mg$$

$$f \leq f_{s \max}$$

$$mg \leq \mu N$$

$$mg \leq \mu ma$$

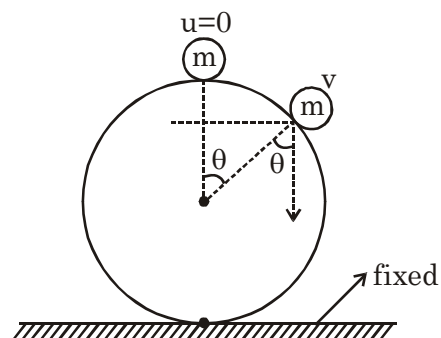
$$a \geq \frac{10}{0.4}$$

$$a \geq 25$$

$$a_{\min} = \frac{25m}{s^2}$$

8. Ans. 1.66 to 1.67

Sol.



$$mg \cos \theta = \frac{mv^2}{r}$$

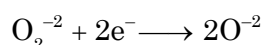
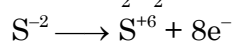
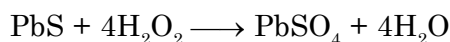
$$mgR(1 - \cos \theta) = \frac{1}{2}mv^2$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(A,B)
2. Ans.(A,B,D)
3. Ans.(B,C,D)
4. Ans.(B,C,D)
5. Ans.(A,D)



6. Ans.(A,C)
7. Ans.(B)
8. Ans. (A)

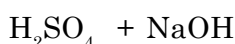
9. Ans.(B)

10. Ans.(C)

SECTION-II

1. Ans.(5.00)
2. Ans(10.8)
3. Ans (4.54)
4. Ans (4.00)
5. Ans. (2.00)
6. Ans. (2.00)

$\therefore$ Total mass of $\text{H}_2\text{SO}_4 = 107.8 \text{ g}$



7. Ans. (22.40)
8. Ans.(30.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A,C,D)

Sol. $\sin A, \sin B, \sin C \rightarrow \text{A.P.}$

$$(A) \frac{a}{2R}, \frac{b}{2R}, \frac{c}{2R} \rightarrow \text{A.P.}$$

$a, b, c \rightarrow \text{A.P.}$

$$(C) \tan \frac{A}{2} \tan \frac{C}{2} = \frac{\Delta}{s(s-a)} \cdot \frac{\Delta}{s(s-c)}$$

$$= \frac{(s-b)}{s} = \frac{\frac{(a+c)+b}{2} - b}{s} = \frac{1}{3}$$

$$(D) \cot \frac{A}{2}, \cot \frac{B}{2}, \cot \frac{C}{2}$$

$$\frac{s(s-a)}{\Delta}, \frac{s(s-b)}{\Delta}, \frac{s(s-c)}{\Delta}$$

$s-a, s-b, s-c \rightarrow \text{A.P.}$

$-a, -b, -c \rightarrow \text{A.P.}$

$a, b, c \rightarrow \text{A.P.}$

2. Ans. (B,D)

Sol. $x = 2, y = 3, z = -1$

$$\begin{vmatrix} 2 & 5 & 3 \\ -1 & 0 & 1 \\ -3 & 2 & 1 \end{vmatrix}$$

$$2(-2) - 5(-1 + 3) + 3(-2)$$

$$-4 - 10 - 6 = -20$$

3. Ans. (A,B)

Sol. $AB = \sqrt{4 + \left(x + \frac{1}{x}\right)^2} = \sqrt{4 + 4} = 2\sqrt{2}$

4. Ans. (A,C)

Sol. $\cos(2x) = 0$ or $\sin x = \cos x$

$$2x = (2n+1)\frac{\pi}{2} \quad \tan x = 1$$

$$x = (2n+1)\frac{\pi}{4} \quad x = \frac{\pi}{4}, \pi + \frac{\pi}{4} \text{ \& } 2\pi + \frac{\pi}{4}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \frac{11\pi}{4}$$

$$\text{Sum} = \frac{\pi}{4} + \frac{3\pi}{4} + \frac{5\pi}{4} + \frac{7\pi}{4} + \frac{9\pi}{4} + \frac{11\pi}{4} = 9\pi$$

5. Ans. (A)

Sol. $\alpha + \beta + \gamma = 1, \alpha\beta + \beta\gamma + \gamma\alpha = 3 \text{ \& } \alpha\beta\gamma = 2$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$= 1 - 2(3) = -5$$

6. Ans. (B)

Sol. $T_r = \frac{1}{2^r} + \frac{(3r^2 + 10r + 6)}{2^r(r^2 + 2r + 2)(r^2 - 2r + 2)}$

$$T_r = \frac{1}{2^r} + \frac{(4r^2 + 8r + 8) - (r^2 - 2r + 2)}{2^r(r^2 + 2r + 2)(r^2 - 2r + 2)}$$

$$T_r = \frac{1}{2^r} + \left(\frac{1}{2^{(r-2)}(r^2 - 2r + 2)} - \frac{1}{2^r(r^2 + 2r + 2)} \right)$$

$$\text{Sum} = \frac{1/4}{\left(1 - \frac{1}{2}\right)} + \left(\frac{1}{2} + \frac{1}{2 \times (5)} \right) = 1 + \frac{1}{10} = \frac{11}{10}$$

7. Ans. (B)

Sol. (P) for determinant $a_{ij} + a_{ji} = 0$
and odd order $\Rightarrow$ value is 0.

(Q) Let $\lambda = |5\sin\theta - 3\cos\theta| \dots(1)$

$5 = (3\sin\theta + 5\cos\theta) \dots(2)$

$(1)^2 + (2)^2 \Rightarrow \lambda^2 + 25 = 25 + 9$

$|\lambda| = 3$

(R) $1 - \cos^2 x - \cos x = \frac{1}{4}$

$\cos^2 x + \cos x - \frac{3}{4} = 0$

$4\cos^2 x + 4\cos x - 3 = 0$

$4\cos^2 x + 6\cos x - 2\cos x - 3 = 0$

$\cos x = \frac{1}{2} \Rightarrow$ Number of solutions are two

(S)
$$\frac{\left(\frac{\Delta}{(s-a)} + \frac{\Delta}{(s-b)}\right)}{\left(2\cos^2 \frac{C}{2}\right)} \times \frac{2\Delta}{abc} = \frac{2\Delta^2}{(s-a)(s-b)ab \cdot \cos^2 \frac{C}{2}}$$
$$= \frac{\Delta^2}{(s-a)(s-b)ab \frac{s(s-c)}{ab}}$$

8. Ans. (A)

Sol. (P) Let $|x-1| = t$

$t^2 - 10t + 7 \leq 0$

$t \leq 5 \cup t \leq 2$

$|x-1| \leq 5 \cup |x-1| \leq 2$

$-4 \leq x \leq 6 \cup -1 \leq x \leq 3$

Sum = $-4 - 3 - 2 - 1 + 0 + 1 + 2 + 3 + 4 + 5 + 6$

(Q) $S = 1 + 2\sin x + 3\sin 2x + \dots \dots(1)$

$(\sin x)S = \sin x + 2\sin^2 x + \dots \dots(2)$

$(1) - (2)$

$\Rightarrow S(1 - \sin x) = 1 + \sin x + \sin^2 x + \dots$

$S = \frac{1}{(1 - \sin x)^2} = 4$

$\Rightarrow (1 - \sin x)^2 = \frac{1}{2}$

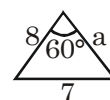
$1 - \sin x = \frac{1}{2} \Rightarrow \sin x = \frac{1}{2}$

$x = n\pi + (-1)^n \frac{\pi}{2};$

number of values are 5

(R) $2B = A + C \Rightarrow B = 60^\circ$

$\cos 60^\circ = \frac{a^2 + 8^2 - 7^2}{2(8)a}$



$a^2 - 8a + 15 = 0 \Rightarrow a = 5, 3$

(S) $1^2 + 2^2 + \dots + 11^2$

$= \frac{11(12)(23)}{6} = 22 \times 23$

9. Ans. (D)

Sol. (P) $\tan^2 60^\circ = 3$

(Q) $-\left(\frac{-b}{2a}\right) = 1 \Rightarrow b = 2a \dots(1)$

$f(1) = a - b - c = -4$

$b + c - a = 4 \dots(2)$

$2b = a + c \dots(3)$

$a = 1, b = 2, c = 3$

$abc = 6$

(R) $\cos^2 \left(x + \frac{\pi}{6}\right) = \cos \left(x + \frac{\pi}{6}\right)$

$\cos \left(x + \frac{\pi}{6}\right) = 0 \text{ or } \cos \left(x + \frac{\pi}{6}\right) = 1$

$x = \frac{\pi}{3}, \frac{-2\pi}{3}, \frac{-\pi}{6} \text{ as } x \in (-\pi, \pi)$

Sum = $\frac{-\pi}{2}; k = -\frac{1}{2}, n = 3$

(S) $\Delta ABC = 3(\Delta ACB) = 9$

10. Ans. (C)

Sol. (P) $\Delta' = \Delta^2 \Rightarrow \frac{\Delta^2}{\Delta'} = 1$

(Q) $(\sqrt{5})^{\log_5 x \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots\right)} = 7$

$(\sqrt{5})^{(\log_5 x) \times 2} = 7 \Rightarrow x = 7$

(R) $(\sin x + \cos x) - 1 = (\sin x + \cos x)^2 - 1 - 2$

$\sin x + \cos x = t \Rightarrow t^2 - t - 2 = 0$

$t = 2 \text{ or } t = -1$

$\sin x + \cos x = 2$ is rejected &

$\sin x + \cos x = -1$

$\Rightarrow \sin x = -1 \text{ or } \cos x = -1$

$$x = \frac{3\pi}{2}, x = \pi, 3\pi$$

total solutions are 3

$$(S) \quad \frac{x^2 + y^2}{2} \geq xy \quad (A.M. \geq G.M.)$$

$$\frac{(y^2 + z^2)}{2} \geq yz$$

$$\frac{(z^2 + x^2)}{2} \geq zx$$

multiply all of them

SECTION-II

1. **Ans. 4.90**

Sol. $L_1 = 5$

$$\frac{25 \tan^2 \theta + 16 \cot^2 \theta}{2} \geq 5 \times 4$$

$$L_2 = 40$$

$$L_3 = 3 + \cos^2 x = 4$$

2. **Ans. 6.00**

$$\text{Sol. } 8 \sin^4 \theta = 3\sqrt{3} \sin \theta \Rightarrow \sin^3 \theta = \frac{3\sqrt{3}}{8}$$

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$\theta = n\pi + (-1)^n \frac{\pi}{3}$$

$$\frac{\pi}{3}, \pi - \frac{\pi}{3}, 2\pi + \frac{\pi}{3}, 3\pi - \frac{\pi}{3}, -2\pi + \frac{\pi}{3}, -2\pi + \frac{2\pi}{3}$$

3. **Ans. 2.75**

Sol. $a = 2\lambda, b = 2\lambda, c = \lambda,$

$$\cos A = \frac{4+1-4}{2(2)} = \frac{1}{4}$$

$$\cos B = \frac{4+1-4}{2} = \frac{1}{4}$$

$$\cos C = \frac{4+4-1}{2(2)(2)} = \frac{7}{8}$$

$$\cos A + \cos B + \cos C = \frac{1}{2} + \frac{7}{8} = \frac{11}{8}$$

4. **Ans. 0.50**

Sol. Let $(x - \alpha)(x - \beta) = 2017$

where a^{2016} and b^{2016} are the roots of quadratic where $\alpha = c^{2016}$ and $\beta = d^{2016}$

$$x^2 - (\alpha + \beta)x + \alpha\beta - 2017 = 0$$

$$(ab)^{2016} = (d)^{2016} - 2017$$

5. **Ans. 0.00**

$$\text{Sol. } \begin{vmatrix} 1 & k & 1 \\ 2 & 3 & -2k \\ 1 & -1 & -1 \end{vmatrix} = 0$$

$$(-3 - 2k) - k(-2 + 2k) + (-5) = 0$$

$$-3 - 2k + 2k - 2k^2 - 5 = 0$$

$$k^2 = -4 ; \text{ Not possible}$$

6. **Ans. 0.50**

$$\text{Sol. } \frac{r}{R} = \frac{\Delta}{s \cdot \frac{abc}{4\Delta}} = \frac{4\Delta^2}{s \cdot abc}$$

$$= \frac{4(s-a)(s-b)(s-c)}{abc}$$

$$s = \frac{15\lambda}{2} \Rightarrow \frac{4\left(\frac{15\lambda}{2} - 4\lambda\right)\left(\frac{15\lambda}{2} - 5\lambda\right)\left(\frac{15\lambda}{2} - 6\lambda\right)}{(4\lambda)(5\lambda)(6\lambda)}$$

$$= \frac{4 \cdot 7 \cdot 5 \cdot 3}{8 \cdot 4 \cdot 5 \cdot 6} = \frac{35}{80} = \frac{7}{16}$$

7. **Ans. 0.25**

Sol. expansion

$$\tan A(\tan B \tan C - 1) - (\tan C - 1) + 1 - \tan B$$

$$(\tan A \tan B \tan C - (\tan A + \tan B + \tan C)) + 2$$

8. **Ans. 1.50**

Sol. $C_2 \rightarrow C_2 - C_1$

$$\begin{vmatrix} 1 & 0 & x \\ p+1 & 0 & p+x \\ 3 & (x-2) & x+2 \end{vmatrix} = 0$$

$$(x-2)p(1-x) = 0$$

$$x = 2, 1$$